CPSC4140
Fall 2026
09/16/2026

Milestone I: Identifying the Problem

The Two-Weekers
Team 05
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Part 1: Exploring the Problem Space

In finding three suitable problems relating to betterment technology, the team emphasized a personal approach, where the idealized technology serves to improve the user's real-world quality of life rather than user experience. As such, finding amendable aspects of users' everyday routines that, in turn, bring some arbitrary gain to their own character became priority. Much of the team's deliberation came from reflecting on their own personal experiences and what parts of their lives they wish they had more complete knowledge, skills, and experience of. After some group discussion and research into existing literature, the team devised three main problem spaces: building a HEALTHIER, THRIFTIER, and SMARTER you.

The problem for the healthier aspect is simple; our health is arguably the most important trait to us, and in a country such as the United States where food consumerism is not as well-regulated, health concerns can begin early and develop rapidly, with weight gain and obesity being one of the first to come to mind given its association with the American identity. However, assisted by the rise in social media popularity, gym habits and lifestyles have become more common than ever before, further popularized by the emergence of digital fitness tracking, which studies suggest is one of the most successful forms of persuasive technology in terms of improving physical health and activity [1]. The issue is not in identifying these healthy habits, but rather maintaining them, as consistency can be an incredibly difficult attribute for some people to develop in strenuous, time-consuming circumstances like being active. As healthy habits can apply to anyone and everyone, the application of this problem space is almost limitless, but a relative level, those who may benefit most are university students with inconsistent, haphazard schedules due to lack of discipline or office workers, many of which are confined to a singular space for multiple hours a day with little activity or movement. The benefit of such a large problem space is the subsequently large solution domain, but with it comes a dense over-saturation in the consumer market. Both analog (routine/workout/hydration journals, calendars, physical reminders) and digital (Apple Health, Samsung Health, fitness apps galore) are readily available to consumers, with each one carrying its own, sometimes unanimous disadvantages. A large recurring problem cited by users stems from irritation in tracking progress, turning what should be a goal users strive for into a chore they must complete. In a particular found example, one user stopped using a hydration tracker app due to their disdain with keeping track of measurements and reaching a set quantitative goal in their drinking, citing, "[i]t stops feeling like self-care and starts feeling like homework" [2]. Furthermore, the disdain for an oversaturated digital market is noted by users as well, making notable mention of the immense number of vibe-coded habit trackers that function nearly identical to one another and all fail because of the lack of identity [3]. This gave the team valuable insight into where to explore the problem space further, particularly in the domain of overcoming tracking and interaction fatigue.

The common problem for the thriftier aspect revolves tightly around the practice of budgeting vs. impulsive buying. Most consumers purchase goods or services without proper thought into whether the purchase is necessary, and some may make purchases without the proper education or context needed to understand if the purchase is valid. Without the proper information or discipline, a budget makes no difference, and much of the issue the team identified can be resolved with providing education and enforcing accountability. For instance, the Financial Consumer Agency of Canada, an agency dedicated to promoting financial literacy among Canadian citizens, conducted a study in 2017 using the Carrot Rewards mobile app to analyze the effects of delivering budgeting advice to users of the app on budgeting knowledge, confidence, and adoption. At the conclusion of the study, it was found that user knowledge rose by 10%, confidence rose by 15%, and adoption rose by 14%, demonstrating that with proper education, individuals may be influenced to better budget their assets [4]. Similar to the first problem space, the idea of financial responsibility is applicable to almost any individual with access to anything of monetary value, and thus the radius of effect is uncontrollably large.However, for simplicity sake, the focus group can be reduced to those fitting any of the team's provided criteria in the milestone I PowerPoint, such as young adults beginning to individually manage their assets, families with home, utility, and grocery bills, and budget/coupon enthusiasts, and environmentally conscious individuals. Likewise, with such a large problem space comes numerous potential yet existing solutions. Analog solutions encapsulate generic records of the aforementioned focus groups, such as written grocery lists and budget requirements, coupon books, and lists of recyclable materials. Digital solutions, such as Rocket Money, Monarch Money, and YNAB, all specialize in budgeting money towards bills and subscriptions, whilst others can specialize in food recording, recipe suggestions, and grocery calculators. Most digital finance solutions serve as permanent replacements to analog solutions of the same type due to convenience and technological ability. For instance, one Reddit user reported struggles managing grocery shopping due to messy and incomplete lists, cascading into extra, unnecessary grocery trips, and thus requesting apps that combine meal and grocery planning [5]. To put it simply, most of the problem domains already have existing solutions, just across multiple apps, and one potential avenue of expansion would be to create an all-in-one amalgamation of budgeting-oriented applications or combine financially adjacent factors present in separate apps into one collective.

Lastly, building a smarter you largely involves "learning learning"; that is, understanding not only WHAT people learn, but HOW they do so as well. Much of the learning humans, particularly students, do extends far beyond the limits of the classroom and notes; accountability in studying, managing due dates, building connections, seizing opportunities, and understanding mistakes and lapses in judgement encapsulate just a few of the extensions of true learning. Although this problem space's population domain is a tad smaller than the previous two due to a narrowed beginning population (high school and university students of any age, most applicable to younger audiences, and professors seeking to understand how their students can learn best), it

remains incredibly large with several branches of potential applicability. Existing analog (paper planners, flashcards, textbooks, and in-person study groups and tutors) and digital (everything aforementioned in the analog technologies, but digitized, like Quizlet, any online textbook system, academic forums and chat groups) technologies capture much of the physical learning/study prerogative, but fall short in planning and accountability. For instance, a study conducted in 2024 analyzed the effects of single-time, minimally invasive digital intervention mechanisms, like advice on decision-making, study sessions, and time management at the conclusion of the given survey, taking five to ten minutes to complete. After analyzing the control vs. study groups, it was found there was little improvement in student-reported procrastination after the intervention, with students struggling to apply the given advice without continual support in academic pursuits [13]. This suggests that advice is not a suitable replacement for continuous support during studying. Existing technologies seek to only improve productivity, which encapsulates a small part of the study process. Furthermore, the overlap in many of the existing technologies creates a capitalistic bubble, where varying softwares upcharge basic features while excluding those that would prove useful. The team found a source from a college student remarking on the current landscape for scheduling/assignment management apps, noting the extensive paywalls, forced usage of artificial intelligence, and required account creation, all the while underdelivering on basic features like a live schedule view with to-do list items and due dates [14]. Therefore, solutions to these aforementioned pitfalls lie in a progressive, continuously adapting study plan/course of action based on courseload, study goals, and personally-set deadlines as opposed to a static reminder or tracker. These solutions implement a motivation/advice approach, which studies suggest is ineffective in student learning, and therefore a new way of encouraging student progress is needed. The previous analysis suggests the best method of doing so is enforcing self-accountability with a route that changes with personally-made progress, preferences, and goals

Part 2a: Problem Overview

- Our problem revolves around money and ways people can save money on many things. We will focus specifically on how people are wasting money on replacing products, which can either be repaired or reused to be economically and environmentally friendly.
- Money is the lifeblood of everyone, even those who are financially stable. Recycling can help in this case as it can be economically and environmentally friendly.
- Existing solutions:
    - Budgeting apps
        - Helps you budget for your financial goals. Can help in saving some money, but can always lead to unnecessary purchases if not budgeted properly.
    - Automated savings
        - Automatically saves money into your savings account for you. It can be difficult to save if income becomes lacking.
    - Price comparison apps
        - Helps compare if the product you are buying can be bought at a cheaper price elsewhere.
    - Secondhand marketplaces
        - Can buy cheaper things from people who have already owned it. Can be questionable quality and cannot be trusted at all times.
    - Food-waste savings
        - Helps buy food that may be expiring soon/surplus in the grocery store. It may be a good savings method, but food won't always last a long time after purchase.
    - Repair technology (iFixit)
        - Helps people repair their products instead of replacing them. Can be helpful for those who are into DIY repairs, but could cost you the product if the repair goes wrong.

Part 2b: Stakeholders and Users

- Retailers
    - Provide people with generic products that can cost a lot or cost little. Consumers have to decide whether the purchase is worth the money or not.
    - A 2026 systematic literature review of 90 peer-reviewed studies identified retailers as important actors in circular business models. Common strategies included reuse, repair, reselling, remanufacturing, refurbishment, recycling, take-back programs, and resale platforms. [6]
- College students
    - Students who are attending and paying for college using their own money.
    - Research on college students has documented significant financial stress and links between financial concerns, academic functioning, and well-being [8].
- Budgeting parents
    - Parents who may want to save as much as possible for their children or a bigger financial goal.
    - Research on families shows that economic pressure affects parents' ability to provide resources and creates significant financial stress within households [7].
- Single parents
    - Parents who may be struggling to financially support their child(ren) with limited financial resources.
    - Stack and Meredith's study of single parents found participants experiencing food and fuel poverty, difficulty paying bills, and sacrifices to provide necessities for their children [9].

Part 2c: Tasks and Context

- Key Tasks
    - Identify the item's problem and determine whether it is safe to use.
    - Find repair information through guides, videos, manufacturers, and professionals.
    - Estimate the cost, time, tools, and skills required for repair.
    - Compare repair with purchasing a new or even a secondhand replacement.
    - Decide whether to repair, replace, continue using, sell, or recycle the item.
    - Evaluate whether the decision produced the better result.
- Important Characteristics
    - The decision usually begins unexpectedly when an item or part breaks.
    - Users may encounter incomplete, conflicting, or extremely technical information.
    - Repair difficulty varies on a multitude of factors like the product itself, the damage done, and the availability of the parts
    - Cost may include parts, tools, labor, shipping, and time
    - Safety risks may limit whether a user should even attempt a repair
    - Users may also need to consider things like convenience, sustainability, and the item's expected lifespan
- The Task Environment
    - Physical: The situation may occur at any place and any time so this may affect the decision depending on the available resources and workspace.
    - Digital: Users can search repair videos, forums, and online marketplaces.
    - Social: Many people in a User's life may have the knowledge or professionalism to repair the device or search for a proper replacement.
    - Organizational: Warranties, store policies, and manufacturer restrictions may limit the available options.
- Hierarchical Task Analysis:
    - 1. Decide whether to repair or replace an item.
        - 1. Identify the item, its damage, and the risks associated with it.
        - 2. Search for likely causes and repair information.
        - 3. Evaluate repair cost, difficulty, time, and expected outcome.
        - 4. Evaluate replacement cost, quality, and expected lifespan.
        - 5. Compare the options set before the user with their needs and priorities.
        - 6. Repair, replace, seek help, or trash the item.
        - 7. Evaluate the outcome

    - Plan 1:
        - Complete Tasks 1 and 2 first, If the item may be unsafe, seek qualified help. Complete Tasks 3 and 4 if both options are possible, then compare them in Task 5. Lastly complete Tasks 6 and 7.

Part 2d: Research Methods and Evidence

- Research Methods
    - We collected 17 peer-feedback responses and grouped common comments in themes that we should use with our research choice.
    - We will survey our fellow classmates as a convenience sample even if they don't technically fit the criteria
    - We reviewed our class readings and read through three ACM Digital Library papers about topics pertaining to our theme
- Gathered Information
    - Eleven classmates explicitly selected the thriftier idea as having the strongest potential, and once again eleven classmates selected that they were most interested in it as well.
    - Classmates warned that budgeting and spending applications already exist, but one person suggested a "repair-versus-replace" idea.
    - Houston et al. found that people repair items for economic, environmental, personal, and social reasons. [10]
    - Rosner and Ames found that access to parts, tools, expertise, money, and community support affects whether a repair is possible. [11]
    - Schaub et al. studied 95 damaged smartphones and found that many people continued using damaged phones because repair and replacement were too expensive. [12]
    - Together these findings show that the repair-versus-replace decisions involve many previously stated factors.
- Potential Future Research
    - Ask classmates about past experiences with broken items and how they decided what to do.
    - Create a survey to identify common decision facts and information sources.
    - Evaluate existing resources such as iFixit, manufacturer support, online forums, and repair shops.

Part 2e: Synthesis and Design Implications

Due to the wide variety of our stakeholders and their vastly different needs for the product we are trying to design. Research does indicate that people are increasingly conscientious about their spending, so the need for assistance is great. We also have found that there are not many products out there that center themselves around repair or reuse which allowed us to find our potential niche solution. However, while each of our stakeholder groups want different things from our product, the tasks expected from it do not change too much. There is also a balance that we have to strike which is that we need to make our product both as accessible as possible, so that all of our stakeholder groups can equally benefit from the product but it also needs to have enough technical detail around the repair or reuse analysis that users will actually get substantive information from it. To that end, we will likely focus our application around it performing on a desktop/virtual platform as that is the most accessible platform in this day and age. If we were to specialize it to be through some kind of physical artifact, we would inevitably lose some accessibility as people may not be able to learn how to handle it.

Part 2f: Peer Feedback and Reflection

Summary

The general consensus from our peers was that our second idea (Building a thriftier/more frugal you) was our most viable problem area for the project. Similarly the concerns everyone held for us were about the same as they all were concerned about the over saturation of our problem areas and whether or not we could make a distinct product. The main strength of the "thriftier you" problem area according to our peers was that it is a problem area particularly of interest to people of our age and occupation. This pointed out that we would have at least one category of stakeholders easily available for interviews or product reviews.

Application

Seeing our peer's feedback was one of the main contributors towards us choosing the "thriftier you" category as it was both the category most people found the strongest but also gave us the best suggestions for what we could do in the category. In order for us to avoid the main concern people had of us making an unoriginal application for a crowded market we plan to make sure that the specific problem we choose will either be original itself or we come up with a unique solution.

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